

Bacterial Wilt - Ralstonia solanacearum

Bacteria called *Ralstonia solanacearum* attack almost 200 plant species in 33 different plant families. This constitutes one of the largest known host ranges for any plant pathogenic bacterium.

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Although to Solanaceae (potato family) contains the greatest number of susceptible species, many other dicot and a few monocot plants are also susceptible. The common name for the diseases this organism causes varies with the host that is attacked. In tobacco, it is called bacterial wilt or Granville wilt (for Granville County, North Carolina, where it was observed as early as 1880) and Moko disease in banana. It is sometimes called southern wilt or southern bacterial

wilt (in the northern hemisphere). This bacterium is noted for diseases caused outdoors in land areas bounded by 45N and 45S latitudes where rainfall averages above 100 cm/year (39 in/year), the average growing season exceeds 6 months, the average winter temperatures are not below 10°C (50°F), the average summer temperatures are not below 21°C (70°F) and the average yearly temperature does not exceed 23°C (72°F).³ It can be moved from such areas into the greenhouse industry in and on plants propagated in those regions and then sold to growers throughout the world. Although the primary location of survival in the environment is in crop and weed hosts, it can also survive in soil. It can be readily spread through the movement of contaminated soil and infected vegetatively propagated plants, in contaminated irrigation water, and on the surfaces of tools (cutting knives) and equipment used to work with the plants, and on soiled clothing.

The bacteria were first named *Bacillus solanacearum*. After several revisions, it was called for many years *Pseudomonas solanacearum*. The latest revision has settled on the name *Ralstonia solanacearum*. It is described as a non-spore forming (spores in bacteria terminology are survival structures rather than units of reproduction as in fungi), Gram negative staining, nitrate-reducing, ammonia-forming, aerobic, rod-shaped (0.5-1.5 µm) bacteria with one polar flagellum. Populations within this genus and species can be further divided into races and biovars based on differing host ranges, biochemical properties, susceptibility to bacteria-infecting viruses (phages), and serological reactions. Race 1 lives in the U.S. where it attacks many floricultural and vegetable bedding plant crops including geraniums (all Pelargonium), Catharanthus, Impatiens, Ageratum, Chrysanthemum, Gerbera, Tagetes, Zinnia, Salvia, Capsicum, Lycopersicon, Nicotiana, Petunia, Solanum melongena (eggplant), Tropaeolum (nasturtium) and Verbena.¹ Race 3 is tropical in distribution and does not occur naturally in North America. Race 3 biovar 2 (R3B2), inadvertently introduced to the U.S. in vegetatively propagated geraniums grown in Central America and Africa, is considered to be a major threat to agriculture in the U.S. because it causes brown rot in potato.² Some of the other known hosts of R3B2 include Pelargonium, tomato, peppers, eggplant, bean, and beet. Weed hosts include black nightshade, climbing nightshade, horsenettle, Jimson weed, purslane, mustards, lambsquarters, and bittergourd. The bacteria can infect through roots and through any fresh wounds. The bacterium can be difficult to work with in the laboratory because it quickly loses pathogenicity and viability in artificial culture.

Symptoms

Above-ground symptoms include wilting of 1-2 leaves on young plants during the heat of the day. Such plants tend to recover at night. On large-leafed plants, only the tissue on one side of the mid-vein may wilt. This is very characteristic of plants such as Nicotiana. Affected leaves turn yellow and remain wilted after a time. The area between leaf veins dies and browns. Usually, the main stem of the affected plants remains upright even though all the leaves may wilt and die.

Internal symptoms include light tan to yellow-brown discoloration of the vascular tissue. Long sections of infected stems reveal dark brown to black streaking in the vascular tissue as the disease progresses. As invasion proceeds, the pith and cortex of the stem become dark brown.

Symptoms in geraniums are very similar to those caused by the bacterial blight pathogen, *Xanthomonas campestris* pv. pelargonii (*Xcp*). However, while *Xcp* can cause leaf spotting, *Ralstonia* does not.¹

Signs Of The Pathogen

Slimy, sticky ooze forms tan-white to brownish beads where the vascular tissue is cut. When an infected stem is cut across and the cut ends held together for a few seconds, a thin thread of ooze can be seen as the cut ends are slowly separated. If one of the cut ends is suspended in a clear container of clean water, bacterial ooze will form a thread in the water.

Management

Growing and propagating from pathogen-free plant material is the main way to avoid problems with *Ralstonia*, regardless of the race and biovar involved. Propagators must use pathogen-free potting soil or other media, establish stock plants that are tested and known to be free of the bacteria, train workers handling the stock plants in methods and procedures that prevent the pathogen from contaminating the potting soil or coming in contact with the stock plants, and then maintaining this system throughout the propagation phase of crop production.

There are no chemicals or biological agents that adequately control these bacteria. Infected plants **must** be discarded as soon as possible.

As is the case with all pathogens carried on vegetatively propagated crops, the purchaser of cuttings or pre-finished plants must isolate all new, incoming plants as if the health of the plants were unknown, even if the plants have been certified as healthy. New plants **must not** be commingled or dispersed among other plants in the greenhouse from other sources. This procedure is crucial because keeping plants originating from one source together allows you to observe those plants as a group, detect any abnormalities within that group, and treat or discard those plants as a group without affecting or damaging plants from other sources. Keeping them together as a group in a defined area of the greenhouse also limits the area that may need to be quarantined, sanitized, or isolated should a pathogen requiring a 'stop sale' (such as *Ralstonia solanacearum*) be found.

References

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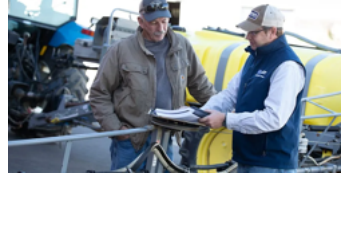
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